

EE 372 Assignment 3: Linear Minimum Mean Square Error Estimation

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1 Introduction

In this assignment, the Linear Minimum Mean Square Error (LMMSE) estimation of a zero mean Wide Sense Stationary (WSS) Random Vector X is implemented based on Chapter 11.3 [1]. We estimate X_1 using subsequent observations $Y(n) = [X_2, \dots, X_{n+1}]^T$ for $n \in \{1, \dots, 6\}$. The objective is to evaluate the impact of statistical accuracy on the estimation performance by comparing theoretical results with data-driven empirical models. The simulations are implemented by using Python [2].

2 Part 1: Autocorrelation Estimation

The true autocorrelation r_k is a normalized sinc function. From Theorem 11.6, for a WSS process, the correlation matrix R_{X_n} must show a Toeplitz structure [1].

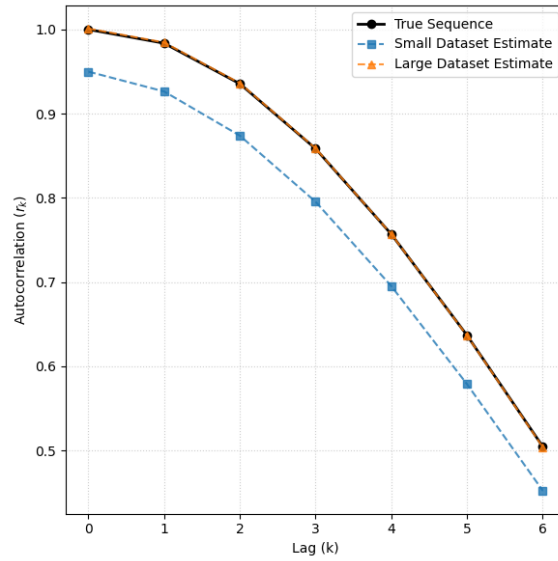


Figure 1: Theoretical r_k vs. Empirical Estimates $\hat{r}_k$ from Small and Large Datasets.

2.1 Convergence of Empirical Statistics

As illustrated in Figure 1, the *large* dataset provides a closer approximation to the theoretical Toeplitz coefficients. Deviations in the *small* dataset highlight that empirical estimates of $R_X[k]$ require a high number of realizations to ensure the stability of the subsequent Wiener-Hopf solution [1].

3 Part 2 & 3: LMMSE Filter Implementation

Following Theorem 11.9, the optimal linear filter coefficients a are derived by solving the discrete-time Wiener-Hopf equations: $R_Y a = R_{XY}$ [1].

- Data-Driven (from Part 2): Weights were calculated using the estimated matrix $\hat{R}_Y$ and vector $\hat{R}_{XY}$ from the large dataset.
- Ideal (from Part 3): Weights were derived using perfect knowledge of the sinc autocorrelation function.

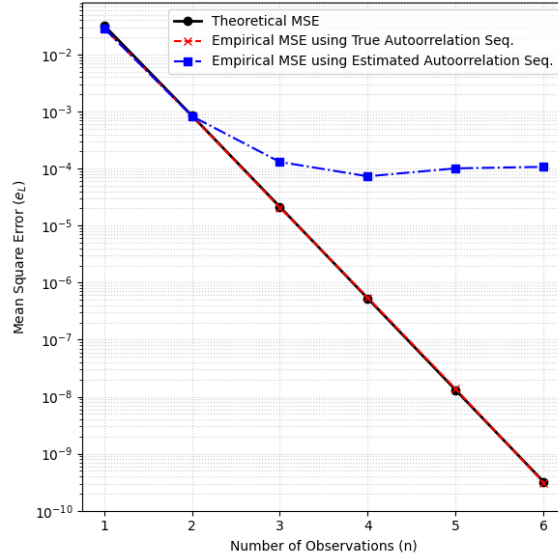


Figure 2: MSE Performance Comparison: Theoretical Bound (e_L^*) vs. Empirical Results.

4 Part 4: Theoretical vs. Empirical Analysis

The theoretical MMSE given by $e_L^*(n) = r_0 - R_{XY}^T R_Y^{-1} R_{XY}$ [1].

- **Ideal Filter Consistency:** The empirical MSE using true statistics (red) aligns perfectly with the theoretical bound (black). This confirms that for a Gaussian-like process, the LMMSE estimator effectively reaches the minimum possible error floor when R_X is known [1].
- **Effect of Mismatched Statistics:** The data-driven estimator (blue) shows a significant performance penalty. Saturating at 10^{-4} despite increasing n . From the principles in Section 11.4, this "floor" is a direct result of using sub-optimal weights a derived from imperfect statistical estimates [1].

5 Conclusion

The study validates that while increasing the filter order n theoretically reduces MSE, the practical performance is constrained by the accuracy of the correlation matrix R_Y . As demonstrated, even a "large" dataset leaves a residual error that prevents empirical estimators from achieving the 10^{-10} precision of the ideal theoretical bound.

References

- [1] Yates, R. D., & Goodman, D. J. (2014). *Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers*. Wiley. [Chapters 11.3-11.4]
- [2] Kenett, R. S., Zacks, S., & Gedeck, P. *Modern Statistics: A Computer-Based Approach with Python*. Springer.